

Physics-Informed Full-Sequence Prediction of Nearshore Wave Time Series

Staged surrogate modeling from offshore forcing with FUNWAVE-TVD cross-shore simulations

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ABSTRACT

Predicting local nearshore wave time series from an offshore forcing record is valuable for coastal engineering because phase-resolving numerical models are accurate but too expensive for rapid scenario screening, repeated design iterations, and large sensitivity studies. This capstone uses one-dimensional cross-shore FUNWAVE-TVD simulations to learn the mapping from one offshore reference time series and a small set of physical descriptors to local surface elevation at shoreward target stations. The final experiment discovered 33,098 usable NetCDF files, selected 5,000 runs for the capstone-default profile, and stationized them into 50,000 offshore-to-local samples. After spin-up trimming, each sequence was resampled to a uniform 0.6 s interval and retained as a full sequence so that travel-time delays and phase relationships were preserved.

Three model stages were evaluated. Stage 1 was a spectral ridge baseline operating in the common forcing-period basis. Stage 2 was a full-sequence conditional temporal convolutional network with FiLM-style static conditioning. Stage 3 added a

differentiable physics layer that applied travel-time shifting and shoaling gain before learning only the residual. On the held-out test split, the physics-informed hybrid achieved the lowest RMSE (0.086), the lowest MAE (0.065), the highest correlation (0.891), the lowest mean absolute lag (0.231 s), and the lowest spectral-band energy errors. Relative to the Stage 2 TCN, the hybrid reduced RMSE by 38.2% and lag error by 76.6%. Depth-binned evaluation showed that mean sample-wise R^2 deteriorated systematically as stations moved into shallower water, but the hybrid remained strong through intermediate and moderately shallow depths and only became unstable in the intermittently wet near-dry regime.

KEYWORDS

Nearshore waves; FUNWAVE-TVD; surrogate modeling; temporal convolutional network; physics-informed machine learning; coastal engineering; full-sequence prediction

1 Introduction

Coastal engineers often need local, phase-resolved wave information at many cross-shore locations, yet high-fidelity numerical models remain computationally expensive when thousands of scenarios must be screened. That tension is especially relevant for beach response studies, structural loading problems, and repeated what-if analyses in which the same offshore forcing must be translated rapidly to many shoreward points. Existing coastal wave tools span a wide range of fidelity. Spectral models such as SWAN are efficient and widely used for coastal wave transformation, while phase-resolving Boussinesq models such as FUNWAVE-TVD capture shoaling, breaking, wetting and drying, and inundation at much higher computational cost (Booij et al., 1999; Shi et al., 2012).

This project addresses the applied machine learning problem of predicting a local wave time series from an offshore reference signal plus a small number of physical descriptors. The intended end users are coastal modelers, engineers, and researchers who already trust phase-resolving simulations but need a fast surrogate for repeated analysis. The data source is a large library of one-dimensional cross-shore FUNWAVE-TVD runs stored as NetCDF files. Each run contains bathymetry, a phase-resolved free-surface field, wet/dry information, and offshore forcing information in the form of wave-component periods, amplitudes, and phases. The surrogate therefore does not replace physics-based simulation in a general sense; it compresses a specific simulator workflow into a much faster station-level predictor.

The central hypothesis of the capstone is that a staged modeling strategy is more defensible than jumping directly to one black-box network. A

simple spectral transfer baseline should establish how much of the problem can be captured with minimal modeling complexity. A full-sequence deep-learning model should then recover phase propagation without sacrificing long-range temporal context. Finally, a physics-informed hybrid should outperform both by embedding travel-time and shoaling structure directly into the prediction pipeline. This hypothesis is consistent with the fact that nearshore wave transformation is shaped not only by amplitude change but also by breaking, nonlinear interactions, and infragravity dynamics that are difficult to reproduce with purely linear transfer assumptions (Battjes & Janssen, 1978; Herbers et al., 1995; Sheremet et al., 2002).

2 Data Summary

The project dataset consists of phase-resolved FUNWAVE-TVD simulations stored as NetCDF files. In plain language, each file contains a cross-shore coordinate, a time axis, a bathymetry profile, the water-surface elevation over time and space, a wet/dry mask, and a set of offshore forcing components defined by period, amplitude, and phase. The capstone pipeline converts those raw fields into station-level supervised learning samples. For each usable run, the notebook selects one offshore reference station and ten shoreward target stations, then stores a full offshore time series, the corresponding local target time series, and static descriptors such as significant wave height, peak period, local depth, station distance from offshore, local slope, path-integrated travel time, and a linear shoaling proxy. Table 1 summarizes the primary variables and their role in the pipeline.

The data required substantial cleaning before model training. The raw files do not share a

uniform temporal resolution, the cross-shore grid length varies between runs, and the wet/dry boundary makes the shallowest portion of each record conditionally valid. The final 5,000-run experiment produced 4,627 unique bathymetry profiles, a median sequence length of 1,755 samples, a median duration of 1,052.4 s, a median offshore depth of 8.16 m, and a median target depth of 4.76 m. The preprocessing notebook trims spin-up using the model-provided propagation time plus a 40 s margin, resamples all sequences to 0.6 s, computes wet fractions from the mask field, and restricts training targets to dynamically valid wet points shoreward of the offshore reference. No preprocessing failures occurred in the final selected subset.

The stationization strategy is deliberate because the project is framed as offshore-to-local transfer rather than full-field image prediction. The final selector chooses the offshore input as the first dynamically valid, sufficiently wet point after the bathymetry begins to change. Shoreward targets are then placed at ten travel-time quantiles between that offshore station and the last dynamically valid wet point, subject to a minimum spacing of 15 m and a target wet-

Table 1

Primary variables used in the station-level surrogate pipeline.

Plain-language variable	Source in the FUNWAVE files	Role in the capstone pipeline
Offshore surface-elevation time series	$\eta(t, x_{\text{off}})$	Primary dynamic input to all three stages.
Local target surface-elevation time series	$\eta(t, x_{\text{target}})$	Supervised target sequence at each shoreward station.
Significant wave height	Hs attribute	Run-level static descriptor used in every stage.
Peak period	Peak Period / Tp attribute	Controls travel time, spectral content, and physics proxies.
Local water depth	Z(X) converted to positive depth	Core local descriptor used in every stage.
Wet/dry validity	mask(t, x)	Defines valid target times and station eligibility.
Wave-component periods, amplitudes, and phases	period, amp, phase	Stage 1 basis representation and offshore forcing summary.
Cross-shore distance and slope	X and $\partial Z / \partial x$	Geometry features used in Stages 2 and 3.

fraction threshold of 0.50. Figure 1 illustrates this process on a representative profile. This design creates a compact but physically meaningful learning problem: one sample is one run-station pair, not one arbitrary sliding window, and full sequences are preserved explicitly to maintain phase lags.

The final training subset intentionally stopped at 5,000 runs so that the full-sequence experiment remained practical on a local workstation or in Google Colab. That subset still produced 50,000 station samples with 35,010 training samples, 7,600 validation samples, and 7,390 test samples, and the split integrity check showed zero overlap in runs or run-station pairs across partitions. Scalar-feature analysis also revealed several strong dependencies, including a 0.97 correlation between offshore distance and travel-time proxy, a -0.94 correlation between depth and travel time, and a 0.85 correlation between significant wave height and target-sequence standard deviation. Those patterns justify both the use of physically motivated features and caution against over-interpreting any single scalar descriptor in isolation.

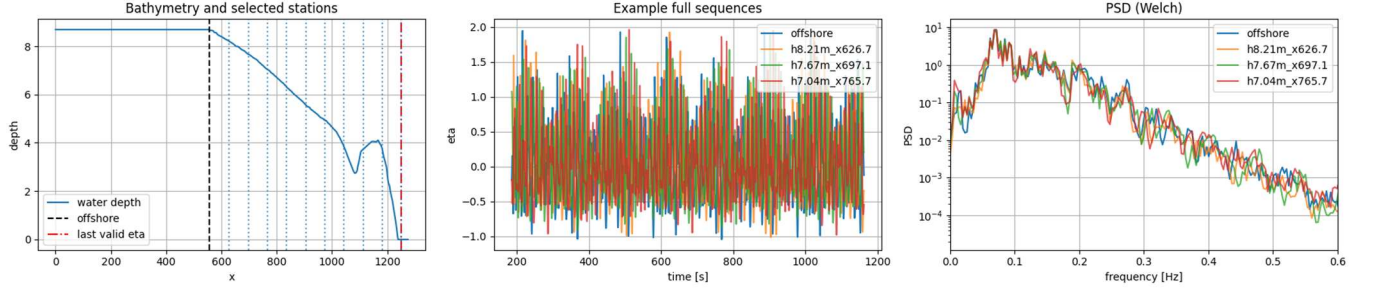


Figure 1

Illustration of offshore reference selection, shoreward target stations, representative full sequences, and spectral content on a representative cross-shore simulation.

3 Literature Review

Conventional coastal-wave prediction is usually built on physics-based numerical models. Spectral action-balance models such as SWAN are a standard choice when the goal is efficient prediction of wave statistics over large coastal domains (Booij et al., 1999). When phase information matters, however, engineers often use phase-resolving models based on Boussinesq-type equations. FUNWAVE-TVD belongs to that class and was developed with a high-order adaptive TVD solver that can represent wave shoaling, breaking, wetting and drying, and coastal inundation (Shi et al., 2012). These traditional models remain the reference standard, but they are expensive enough that surrogate modeling is attractive for repeated analysis.

The nearshore environment also presents a modeling challenge that is particularly relevant to this project. Depth-induced breaking redistributes energy and changes the mean water level in ways that simple linear transfer models do not reproduce well (Battjes & Janssen, 1978). In addition, infragravity motions are generated and amplified as wave groups propagate shoreward, especially across the shoaling zone, and those low-frequency components are strongly affected by dissipation and shoreline reflection (Herbers et al., 1995; Sheremet et al., 2002). This makes the task more difficult than predicting a stationary

offshore signal at a shifted time lag. A surrogate model must preserve phase while still adapting to nonlinear nearshore transformation.

For sequence learning, temporal convolutional networks are a strong alternative to recurrent architectures because they offer large receptive fields, stable optimization, and strong empirical performance on sequence tasks (Bai et al., 2018). They are especially attractive here because the capstone deliberately avoids temporal pooling and sliding windows; the model must accept a full offshore sequence and return a full local sequence without losing temporal alignment. Static conditioning is also important because the same offshore record can lead to different local behavior depending on depth, bathymetry, and propagation distance. Feature-wise linear modulation, or FiLM, is a lightweight conditioning mechanism that applies feature-wise affine transformations based on auxiliary inputs and has proven effective in conditional neural architectures (Perez et al., 2018).

Physics-informed machine learning provides a second major theme in the literature. In the broad sense introduced by Raissi et al. (2019), physics-informed neural networks incorporate governing-equation structure or physical constraints into training. For this capstone, a full PDE PINN would be heavier than necessary because the simulator has already generated target data. A

more practical choice is a hybrid residual architecture in which a simple physics layer handles the most interpretable part of the mapping and the neural network learns only the correction. That design preserves physical intuition while remaining computationally tractable on local hardware. Looking beyond the current station-based formulation, operator-learning methods such as the Fourier Neural Operator are also relevant future candidates for full-field surrogate modeling (Li et al., 2021).

4 Methodology

The pipeline used in the notebook is shown in Figure 2. Raw NetCDF files are discovered recursively, lightweight metadata are extracted, and each usable run is converted into one offshore reference sequence plus a set of ten shoreward station targets. The implementation is intentionally local-PC and Colab friendly: it supports automatic path resolution, optional archive extraction, cached preprocessing artifacts, and either RAM-based or lazy loading depending on available memory. That engineering layer matters because a capstone project is evaluated not only on model choice but also on whether the full training and inference workflow can be reproduced in a practical environment.

Stage 1 is the simplest baseline. Because the FUNWAVE runs share a common forcing-period basis, the pipeline projects both the offshore sequence and the local target sequence onto cosine and sine coefficients at those periods. A multi-output ridge regressor is then trained from offshore spectral coefficients plus the strict-core static features significant wave height (H_s), peak period (T_p), and local depth to the local cosine and sine coefficients (Hoerl & Kennard, 1970). The final local sequence is reconstructed from the predicted coefficients. This design is fast, interpretable, and appropriate as a lower-

complexity baseline, but it is not expected to preserve detailed phase evolution in the shallowest nearshore region.

Stage 2 is the first full-sequence deep-learning model. It uses a conditional temporal convolutional network implemented in PyTorch (Paszke et al., 2019) with one-dimensional residual convolution blocks, dilation factors from 1 to 128, kernel size 5, hidden width 32, dropout of 0.06, and no temporal pooling. The dynamic input channels are the offshore sequence, its Hilbert-envelope magnitude, and its temporal derivative. Static conditioning uses H_s , T_p , local depth, offshore-to-target distance, and local bathymetric slope. FiLM-style modulation injects the static features into each residual block (Perez et al., 2018). The model is therefore full sequence in, full sequence out, with variable-length padding and masking but no sliding windows.

Stage 3 extends the TCN into a physics-informed hybrid. Before the residual network is applied, a differentiable physics layer shifts the offshore sequence by the path-integrated travel time τ and scales it by a linear shoaling gain computed from the cached path features. The layer is not fixed: it learns only small corrections to τ and gain, capped at $\pm 25\%$ for travel time and $\pm 20\%$ for shoaling gain, while a residual TCN predicts the remaining mismatch. The hybrid uses the same 32-channel TCN backbone as Stage 2, but with additional dynamic channels for the delay trace and deterministic physics baseline plus two additional static features, τ_{path} and $\text{shoaling_gain_path}$. The loss combines a masked Huber data term, masked correlation loss, infragravity-band and sea-swell-band spectral penalties, and a regularizer on the learned physics corrections.

Training and evaluation were performed at the run level to prevent data leakage.

GroupShuffleSplit created approximately 70% training, 15% validation, and 15% test partitions such that stations from the same simulation could not appear in multiple splits. Under the capstone-default profile, Stage 1 searched a ridge penalty grid from 10^{-4} to 10^1 and selected $\alpha = 10.0$, yielding a validation coefficient-space MSE of

7.93×10^{-4} . Stage 2 trained for up to 20 epochs with AdamW, gradient clipping, and early stopping patience of 5; Stage 3 trained for up to 25 epochs with patience of 6. Power spectral density comparisons in the results section use Welch estimates (Welch, 1967).

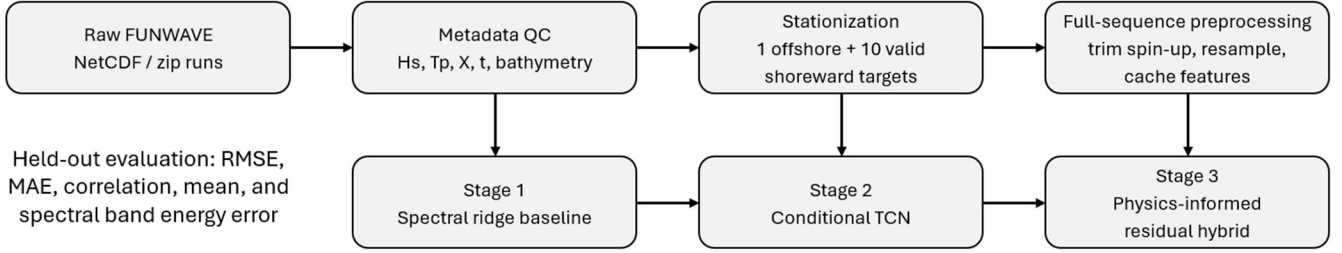


Figure 2

High-level workflow from raw FUNWAVE runs to stationized samples, staged surrogate models, and held-out evaluation.

5 Results

The final results represent a capstone-scale experiment on 5,000 runs sampled from a much larger discoverable corpus. Even at that reduced scale, the problem is substantial because the full-sequence formulation preserves long temporal context and each run contributes ten target stations. The preprocessing step generated 50,000 station samples. Training behavior was healthy for the two neural-network stages, as shown in Figure 3. The Stage 2 TCN reached its best validation loss of 0.2362 at epoch 15, whereas the Stage 3 hybrid reached its best validation loss of 0.1137 at epoch 23. Neither curve showed the late-epoch divergence that would indicate catastrophic overfitting.

Table 2 summarizes the held-out test performance. The pure physics baseline, which only applied a travel-time shift and shoaling gain, was not competitive on its own. The Stage 1 ridge baseline improved RMSE substantially relative to the physics-only model, but it still produced large lag errors and poor waveform agreement. This is an important result rather than

a failure: it shows that amplitude-level spectral transfer is not sufficient when the evaluation target is the full local waveform. Stage 2 was the first model to produce clearly useful phase-sensitive predictions, reaching $\text{RMSE} = 0.140$, $\text{correlation} = 0.684$, and a mean absolute lag = 0.99 s.

Stage 3 delivered the strongest practical result. On the held-out test split it reached $\text{RMSE} = 0.086$, $\text{MAE} = 0.065$, $\text{correlation} = 0.891$, and mean absolute lag = 0.231 s. Relative to Stage 2, the hybrid reduced RMSE by 38.2%, reduced lag error by 76.6%, reduced infragravity-band relative error by 34.8%, and reduced sea-swell-band relative error by 58.1%.

It should be noted that depth-binned analysis clarifies that R^2 issue and is one of the most important results of the project. Figure 4 shows that the Stage 3 hybrid was strongest across depth bins in RMSE, correlation, and lag error, while mean sample-wise R^2 deteriorated progressively as depth decreased. For stations deeper than 6 m it achieved $\text{RMSE} = 0.062$, $\text{correlation} = 0.961$, and $R^2 = 0.925$. The model still worked well through most of the profile and remained

practically useful into very shallow water: even in the most difficult ≤ 1 m bin, it reduced RMSE to 0.106 and lag error to 0.97 s while improving correlation to 0.664, although mean R^2 remained strongly negative there because the variance collapsed. Representative output in Figure 5 reinforces the aggregate metrics: the Stage 3 hybrid aligns the waveform more closely in both the full record and the zoomed segment, while its power spectral density remains most consistent with the target across the dominant sea-swell band and the low-frequency tail.

From a capstone perspective, these results create a coherent narrative across all three stages. Stage 1 provides a fast, interpretable baseline and

documents the limitations of a purely spectral transfer view. Stage 2 demonstrates that a full-sequence TCN can learn meaningful propagation and local transformation without resorting to windowing. Stage 3 shows that hybrid physics-informed design produces the strongest practical model. The main remaining limitation is that the final results are still based on a computationally selected 5,000-run subset rather than the full 33,098-file corpus. A production-grade extension should therefore preserve the same evaluation protocol while training on a larger fraction of the library and adding additional diagnostics in the shallow near-dry regime.

Table 2

Held-out test performance for the final 5,000-run capstone experiment. Lower RMSE and lag are better; higher correlation is better.

Model	RMSE	Correlation	Mean lag [s]
Physics baseline	0.423	0.028	4.84
Stage 1 spectral ridge	0.255	0.136	20.32
Stage 2 full-sequence TCN	0.140	0.684	0.99
Stage 3 physics-informed hybrid	0.086	0.891	0.23

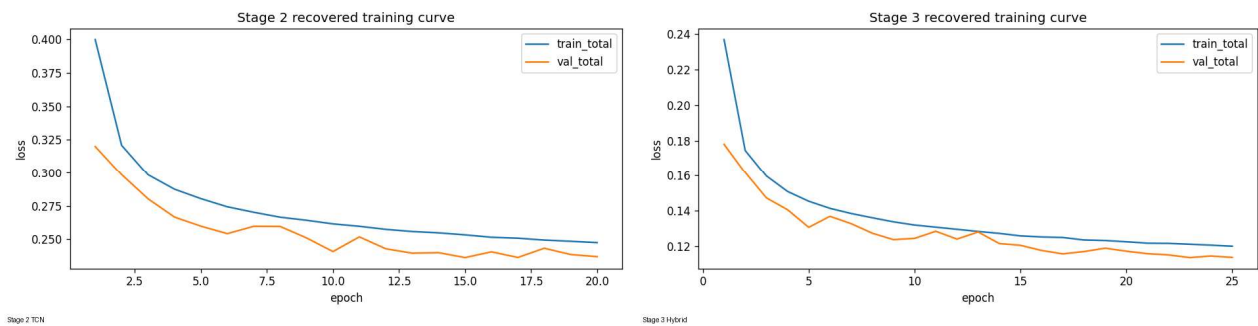


Figure 3

Stage 2 and Stage 3 training histories recovered from the final notebook. The hybrid reached the lower validation loss and continued improving later into training.

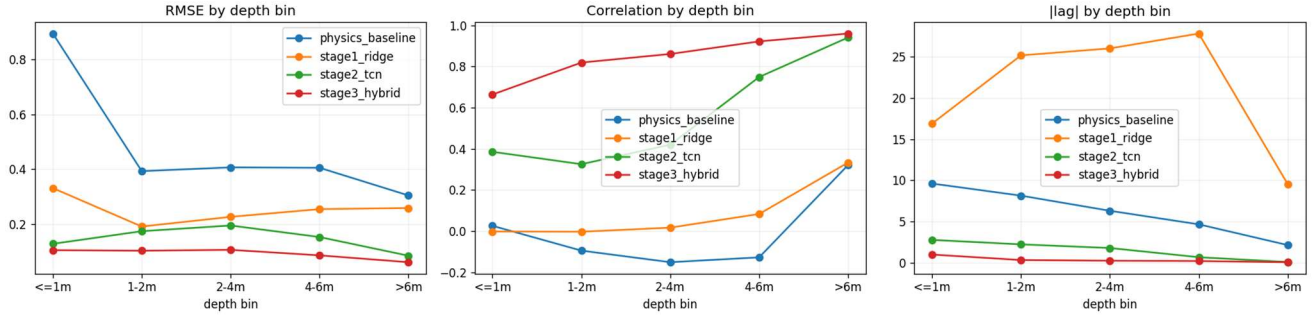


Figure 4

Depth-binned held-out performance. The hybrid remained strongest across bins and showed that mean sample-wise R^2 deteriorated mainly as the model approached the shallowest near-dry regime; practical waveform skill remained strong until very shallow water and collapsed only in the lowest-depth bin.

Recovered held-out prediction example

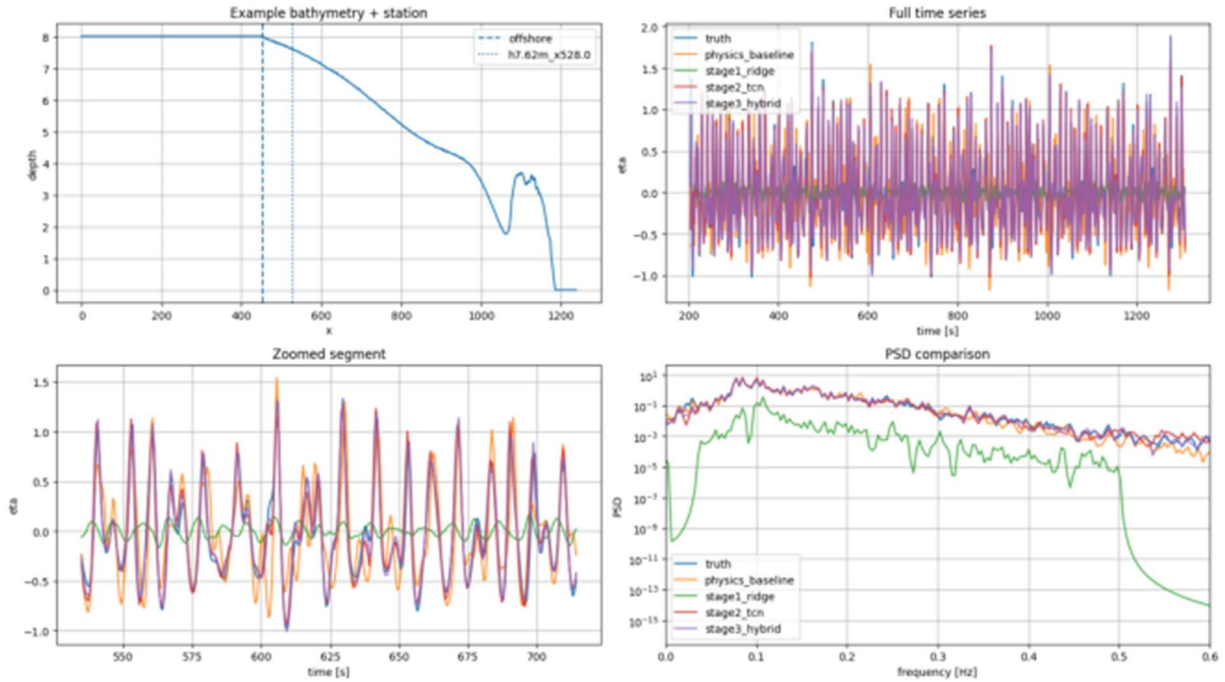


Figure 5

Representative held-out station prediction. The hybrid aligns best with the measured waveform in the full record, the zoomed segment, and the frequency-domain comparison.

6 Conclusion

The goal of this capstone was to build a practical, staged surrogate model that predicts local nearshore wave time series from an offshore forcing record and a small set of physical descriptors. The final experiment supports the

original hypothesis. A simple spectral model was useful as a baseline but insufficient for full-waveform prediction. A full-sequence conditional TCN recovered meaningful phase propagation, and a physics-informed hybrid improved the result substantially by embedding differentiable

travel-time and shoaling structure into the network.

The most significant project result is the Stage 3 test performance on the final held-out split: RMSE = 0.086, MAE = 0.065, correlation = 0.891, mean absolute lag = 0.231 s, and the lowest relative spectral-band errors in both the infragravity and sea-swell ranges. Equally important is the depth-regime finding that mean sample-wise R^2 deteriorated as the target stations moved into shallower water, yet the model continued to work well until the very shallow near-dry regime, where variance-normalized R^2 became unstable. The pipeline is also reproducible on a local PC or in Colab through cached preprocessing, adaptive loading, and staged model complexity. If the project continues, the next steps are clear: train on a larger fraction of the 33,098-run corpus, add more diagnostics for the ≤ 1 m regime, test stricter generalization splits by bathymetry profile, and evaluate whether operator-learning approaches can extend the method from station prediction to full cross-shore fields (Li et al., 2021).

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